CONNECTOMES FOR AI

The blueprint for better AI is wired into the brain.

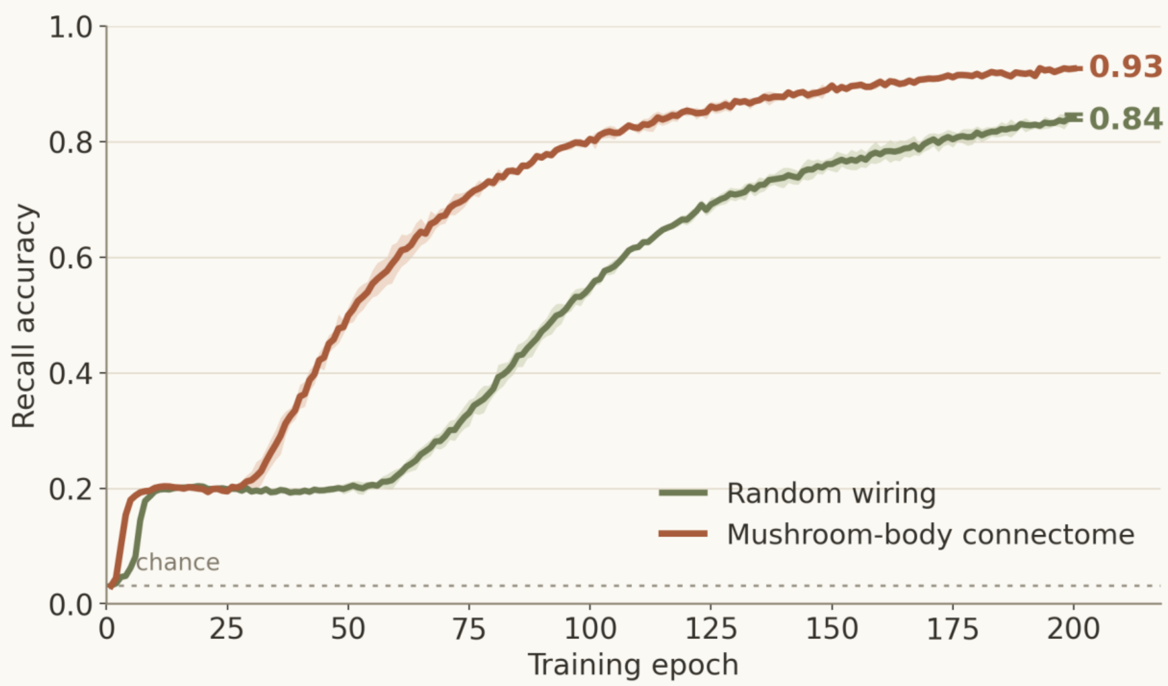
Some problems that are effortless for brains still challenge the best AI models — recalling, recognizing, learning from a handful of examples. The brain's edge is its architecture, refined over hundreds of millions of years. We read that wiring diagram— the connectome — and built it directly into recurrent neural networks to ask if they could learn brain-native tasks.

Collect Connectome Data → **Brain-Derived Models** → **Alignment on Brain-Native Tasks**

FINDING 1 · THE CONNECTOME WIRING IS OPTIMIZED FOR ITS TASK

The connectome learns faster and remembers more than artificial models

The fly mushroom body performs associative learning by binding sensory cues to outcomes. We asked how this circuit, when packaged as a recurrent neural network, compares to size-matched sparse models on the task it evolved for. The connectome learned associative pairings more accurately and quickly than randomly wired circuits.



FlyWire mushroom body, 14,025 neurons, 2 seeds (shaded band = range). A stricter degree-matched control does worse still (0.77); the connectome's lead is statistically significant (p = 0.048).

0.93 vs 0.84

final recall accuracy — connectome vs random wiring

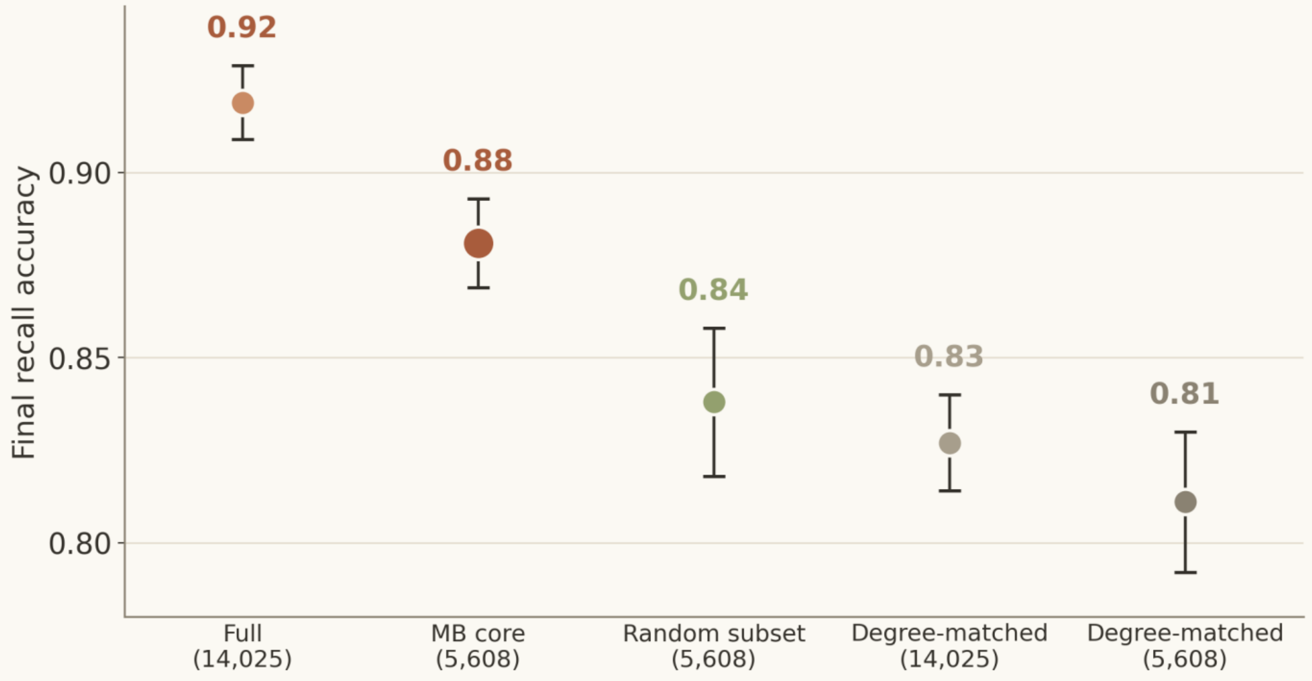
~1.7× less time

wall-clock training to reach 80% recall (same-size networks)

FINDING 2 · SMALLER MODELS THAT TRAIN FASTER

Biologically-informed pruning produces smaller models at minimal cost

Based on knowledge of the mushroom body circuit, we pruned the original neuron count from 14,025 to 5,608 – keeping only the cells that are biologically important to learning. Despite a 2.5x reduction in parameter count, task performance marginally moved (0.92 → 0.88), training ran 2.5× faster, and the pruned circuit beat every random circuit with more parameters.



MQAR, lr 1e-3, 20 seeds (degree-matched controls n=14–17); error bars = ±1 SD. The MB core beats every control graph (rank 0 of N).

0.92 → 0.88

recall after removing 60% of the neurons — keeps ~96% of full accuracy after the same number of training runs

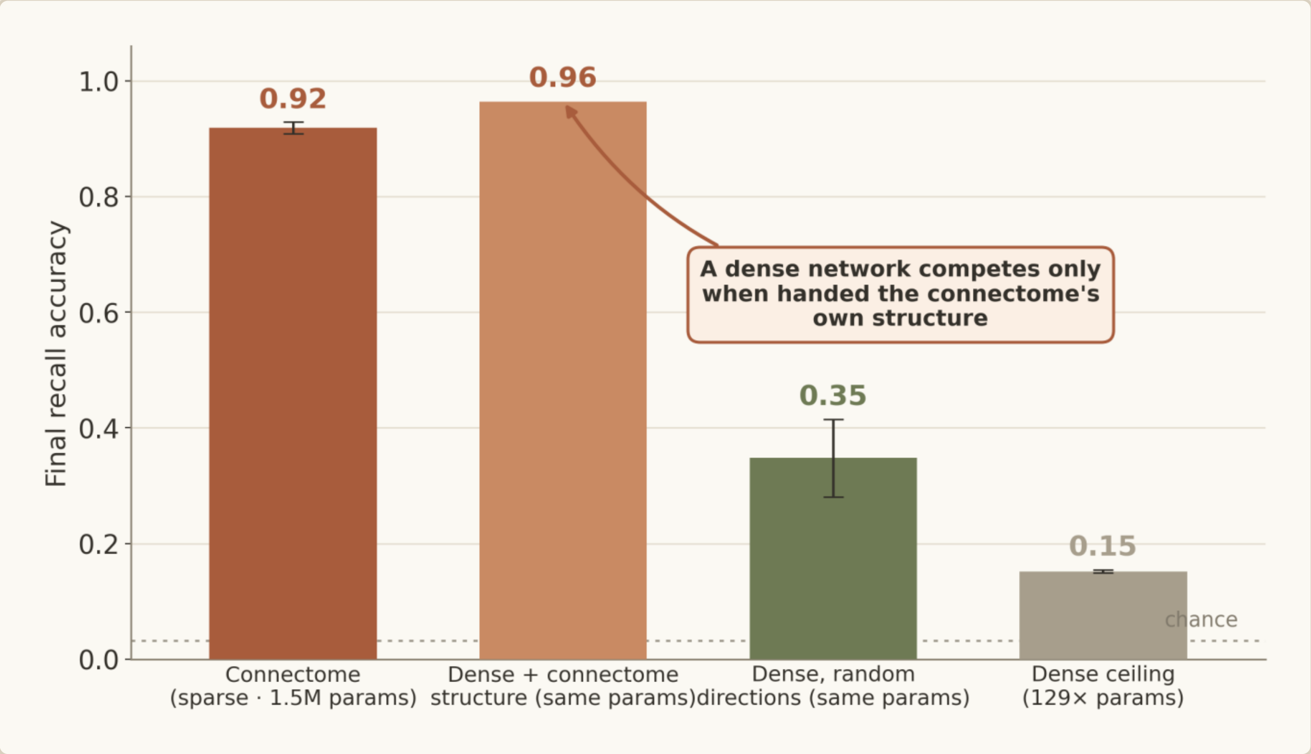
2.5× faster

wall-clock training time – suggesting better performance in less time.

FINDING 3 · CONNECTOME WIRING vs RAW PARAMETERS

It's the structure, not the parameter count

To test if the connectome's performance can be matched by simply boosting parameter count, we compared it to fully-dense recurrent networks. Despite having up to 129x more parameters, the naïve dense approach yields minimal learning. Only when handed the connectome's own structure do dense networks compete.



MQAR, 14,025-neuron network, 20 seeds; error bars = ±1 SD. "Dense + connectome structure" = a dense net carrying the connectome's directions (Exp 2). Matched = equal trainable params (~1.5M).

0.15 recall with 129× params

a dense net with 129× the parameters still fails to learn

~2.8× faster

the sparse connectome also trains quicker than matched dense nets

THE BET

Collecting more connectome data is the path to a new class of AI that solves what brains find easy and machines find hard.

Fruit fly → mouse → human